

BCG treatment for bladder cancer

BCG is a type of immunotherapy drug. It is used to treat some non-muscle-invasive bladder cancers.

On this page

- [What is BCG treatment?](#)
- [When is BCG treatment used?](#)
- [Having BCG treatment](#)
- [Side effects of BCG treatment](#)
- [About our information](#)
- [How we can help](#)

What is BCG treatment?

BCG is short for Bacillus Calmette Guérin. It is a type of [immunotherapy](#) drug. Immunotherapy drugs encourage the body's immune system to fight cancer cells.

Some people may know BCG as a vaccine used to prevent tuberculosis (TB). But it is also a main treatment for non-muscle-invasive [bladder cancer](#).

For bladder cancer, BCG is given directly into the bladder (intravesical). This can make the bladder react in a way that makes the immune system get rid of cancer cells.

BCG availability

Recently there have been some difficulties with the availability of BCG. Your doctor or nurse will explain if this is likely to affect your treatment.

When is BCG treatment used?

You may have BCG directly into your bladder after a [transurethral resection of a bladder tumour \(TURBT\)](#). There is usually at least 2 weeks between the TURBT and the start of BCG treatment. This is to give your bladder enough time to heal from the surgery.

Doctors usually suggest this treatment if you have a [high-risk bladder cancer](#). It is sometimes used if you have an intermediate-risk bladder cancer. This treatment helps prevent the cancer from coming back in the bladder lining. It also reduces the risk of the cancer becoming muscle-invasive.

You usually have BCG treatment once a week for 6 weeks. This is called the induction course. Some people then have more BCG treatments. This is called maintenance treatment.

Your doctor will explain how many treatments you will have.

Having BCG treatment

You have this treatment in the hospital outpatient department. Your nurse or doctor will give you advice on preparing for your treatment. For example, they will ask you to limit the amount of fluids you drink 4 to 6 hours before BCG treatment. This can:

- make it easier for you to keep the BCG in your bladder for the required time – this stops you needing to pass urine (pee) too early
- increase the concentration of the BCG in your bladder.

Tell your doctor if you take any medications. People who take water tablets should take them after the treatment. Water tablets are also called diuretics.

When you arrive for your appointment you will be asked to pass urine. Your nurse will check this for infection. You will not be given BCG if you are feeling unwell or have a urine infection.

To have the treatment, a nurse passes a tube called a catheter through the urethra and into the bladder. They then slowly put the BCG through the catheter directly into the bladder.

Once the drug is in the bladder, the nurse usually takes the catheter out. They ask you not to pass urine for 2 hours. This gives the BCG time to work.

You can get up and walk around during this time. Your bladder may feel full and a bit uncomfortable but you can go to the toilet as soon as the treatment is finished.

Some hospitals will let you go home after the BCG has been put into your bladder and the catheter has been taken out. You are still asked not to pass urine for 2 hours. Once the 2 hours have passed you can go to the toilet at home.

Sometimes, the nurse leaves the catheter in and clamps it to keep the BCG in your bladder. When the treatment is over, the nurse removes the clamp and the BCG drains from your bladder into a urine bag. Your nurse then takes the catheter out.

After treatment

It is important to remember that BCG is a live vaccine and other people should not be exposed to it. The nurse will ask you to take certain precautions for 6 hours after treatment. This will protect you and others from coming into contact with the BCG.

You will be asked to do the following:

- Sit down to pass urine. This avoids splashing urine on the toilet seat.
- Wash your hands and skin in and around your genital area with soap and water after you pass urine. This cleans any drops of BCG that may have splashed on to your skin
- Put undiluted bleach into the toilet bowl after you have passed urine and leave it for 15 minutes before flushing.

For the first few days after each treatment, try to drink 2 to 3 litres (3½ to 5 pints) of fluid a day. This helps to flush the drug out of your bladder.

Talk to your nurse if you are worried. They will tell you what you need to do after treatment.

Sex and contraception

You are usually advised not to have sex for at least 48 hours after the treatment as it may be uncomfortable.

Small amounts of BCG can get into your body fluids. This includes fluid made in the vagina and the fluid that contains sperm. To protect any partners, your cancer doctor may advise that for a week after this treatment you use:

- a condom for vaginal or anal sex
- a condom or a latex barrier such as a dental dam for oral sex.

Your cancer doctor or specialist nurse can give you more information about this.

Doctors do not yet know how BCG may affect an unborn baby. They will recommend you do not [become pregnant](#) or [make someone pregnant](#) while having it. You should use effective contraception during treatment. Your doctor or specialist nurse can give you more information about this.

Side effects of BCG treatment

You may have some side effects after your treatment. These include:

- needing to pass urine (pee) often
- pain when you pass urine
- blood in the urine
- flu-like symptoms, such as tiredness, general aching and a raised temperature.

These effects should get better in 48 to 72 hours. Tell your doctor straight away if they do not. Taking painkillers can help.

Rare side effects include:

- a continuing high temperature (fever)
- vomiting
- skin rash
- a cough
- a condition called reactive arthritis – this can cause symptoms such as painful joints, symptoms of a urinary tract infection and red and sore eyes.

If you have any of these or feel generally unwell, tell your doctor straight away. It could be a sign of a more serious infection due to BCG that needs treatment straight away. If this happens, you will be treated with antibiotic drugs used to treat tuberculosis (TB).

About our information

References



Below is a sample of the sources used in our bladder cancer information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

Mottet N, Bellmunt J, Briers E, et al. Non-muscle-invasive bladder cancer (TaT1 and CIS). European Association of Urology (Internet), 2021. Available from uroweb.org/guideline/non-muscle-invasive-bladder-cancer (accessed September 2021).

Witjes JA, Bruins HM, Cathomas R, et al. Muscle-invasive and metastatic bladder cancer. European Association of Urology (Internet), 2021, Available from uroweb.org/guideline/bladder-cancer-muscle-invasive-and-metastatic (accessed September 2021).

Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, [Dr Ursula McGovern](#), Consultant Medical Oncologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

Date reviewed

Reviewed: 01 November 2022 | Next review: 01 November 2025

⚠ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

[Get help](#)